Under the surface.

What we built, how it works, and where this Android APK can run.

NATIVE JAVA
No WebView UI

ANDROID API 26+
Target API 30

LANDSCAPE TV UI
1080p on tested Stick

ADB SIDELOAD
Same-key app + helper

01

Native application

Java Activity with Android Views, Canvas drawing and hardware acceleration. No Compose, web UI or Google Play Services dependency.

uk.michael.calmhome is a single-task, landscape APK. PackageManager discovers enabled, exported launcher activities; package icons/banners are read locally and fitted to 16:9 tiles.

02

Remote interaction & state

Six-column grid; 5.5% focus enlargement over 180 ms; thin rounded gloss outline. Artwork crossfades in 220 ms and grid transitions run for 260 ms.

A 650 ms Select hold opens Move/Remove. SharedPreferences saves order, hidden tiles and focus. Back cancels moves; backgrounding rolls back unfinished changes.

03

Memory & background work

Separate feed, artwork and local-hero workers. LRU budgets: app icons 12 MiB, shared media 12 MiB, brand heroes about 17.6 MiB.

Hero decode cap: 1920 x 1080; station previews: 640 x 360. Artwork reads are capped at 4 MiB, with network timeouts and a read deadline. Grid/paused states defer invisible work; memory pressure clears caches.

04

Top Shelf & favourites

Reads Frame's content provider and available Android TV shared-programme tables. Select routes to source-app details; Play/Pause can request playback. Unsupported apps use brand artwork.

Opt-in /favourites providers supply station title, artwork and intent. Up to 40 previews per provider, at 4/3 app-tile size. Links are restricted to the publishing app. Sounds Like integration is pending.

05

Home redirect & permissions

A separate, modified Home on Fire accessibility service redirects Home/wake. Event batching: 0 ms; early wake check: 80 ms, with a 500 ms fallback. Amazon Home may still flash.

Launcher access includes package discovery, network state and shared TV listings, subject to Fire OS restrictions. Frame reads and the Fire OS escape use same-signature permissions. No root, system-launcher removal or silent system-service shutdown.

06

Build, deployment & verification

JDK + Android SDK platform 30 + build-tools 35.0.0. Plain aapt / javac / d8 / zipalign / apksigner build pipeline; no Gradle dependency.

Sign launcher and helper with the same local key; retain it for updates. Install via ADB, then enable the helper on the TV. 117 device checks passed; not a multi-day soak or a 60 fps guarantee.

DEVICE COMPATIBILITY [1, 2]

The OS matters more than the model name.

TESTED
Fire TV Stick 4K, 2nd gen (2023)
AFTKM. Fire OS 8 / Android 11, API 30. UI: 1920 x 1080; HDMI: 4K at 59.94 Hz.

EXPECTED - NOT TESTED
Other Android-based Fire OS 8 devices
Examples: Stick 4K Max 2nd gen (2023), 4K Plus (2025), and Fire OS 8 televisions.

EXPECTED - NOT TESTED
Android-based Fire OS 7 devices
API 28: Stick 3rd gen/Lite, HD (2024), 4K Max 1st gen and Cube 2nd/3rd gen. Helper timing may differ.

NEEDS DEVICE VALIDATION
Android TV / Google TV, Android 8+
Meets the APK API floor. Untested; default-home selection and Fire OS-specific controls need validation/adaptation.

NEEDS DEVICE VALIDATION
Fire OS 14 / 16
API level is sufficient, but neither these devices nor the Home helper behaviour has been tested.

NOT SUPPORTED BY THIS APK
Fire OS 5 / 6
API 22 / 25 are below minSdk 26. Includes the original Fire TV Stick 4K (2018).

NOT SUPPORTED BY THIS APK
Vega OS Fire TV devices
4K Select (2025), Stick HD (2026), Stick 4K 3rd gen (2026). A separate implementation is required.

NOT A NATIVE TV APP
Other TV operating systems
Samsung Tizen, LG webOS, Apple tvOS and this Panasonic TX-49FX555B. Use a compatible Stick over HDMI.

Only AFTKM is verified. Other eligibility is inferred from API requirements; firmware can change Home redirection.

[1] Amazon device / OS identification

[2] Amazon Fire OS overview

[3] Source, build and stability report